

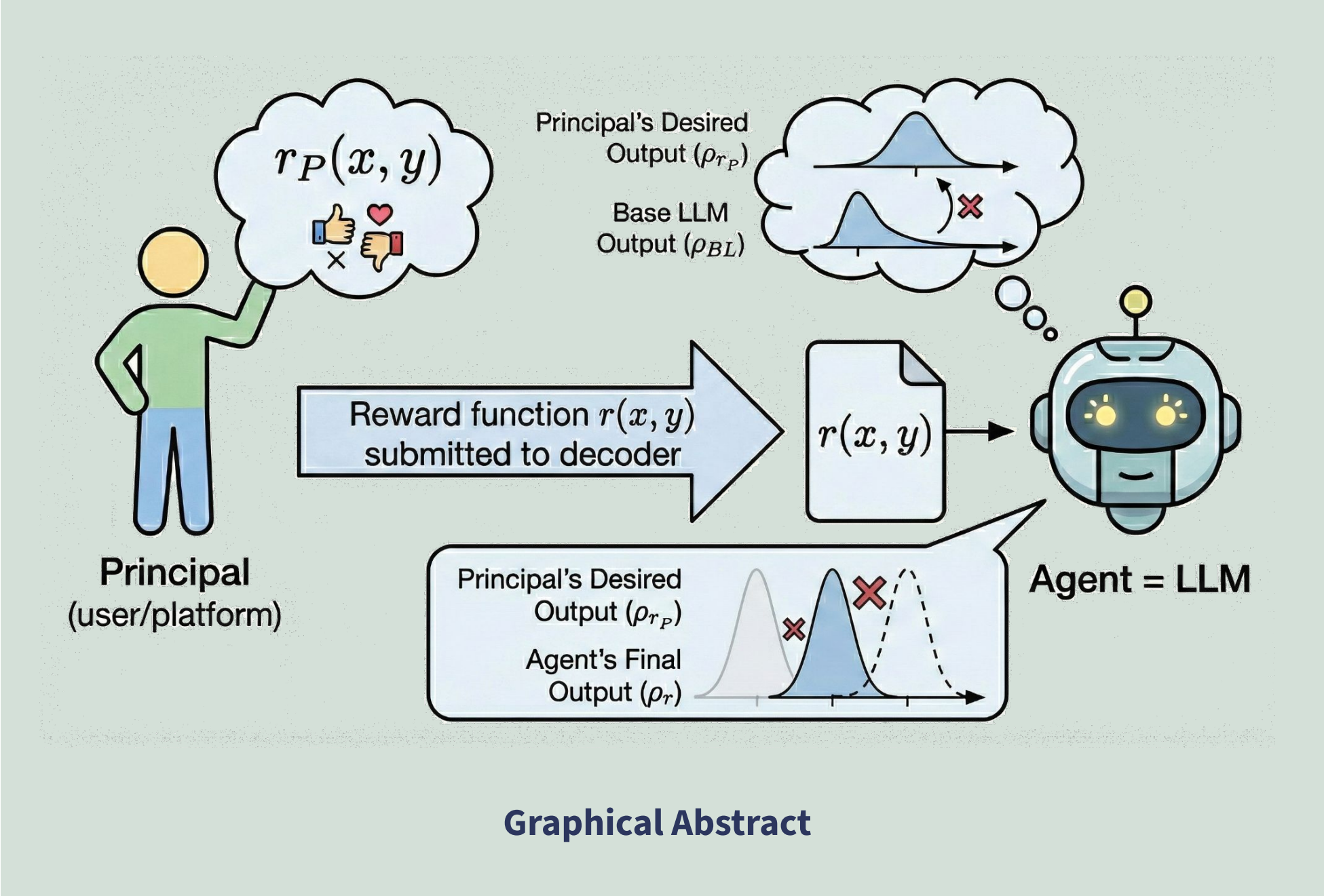
Mitigating Preference Conflict in Controlled Decoding: A Principal-Agent Perspective

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Our Contributions

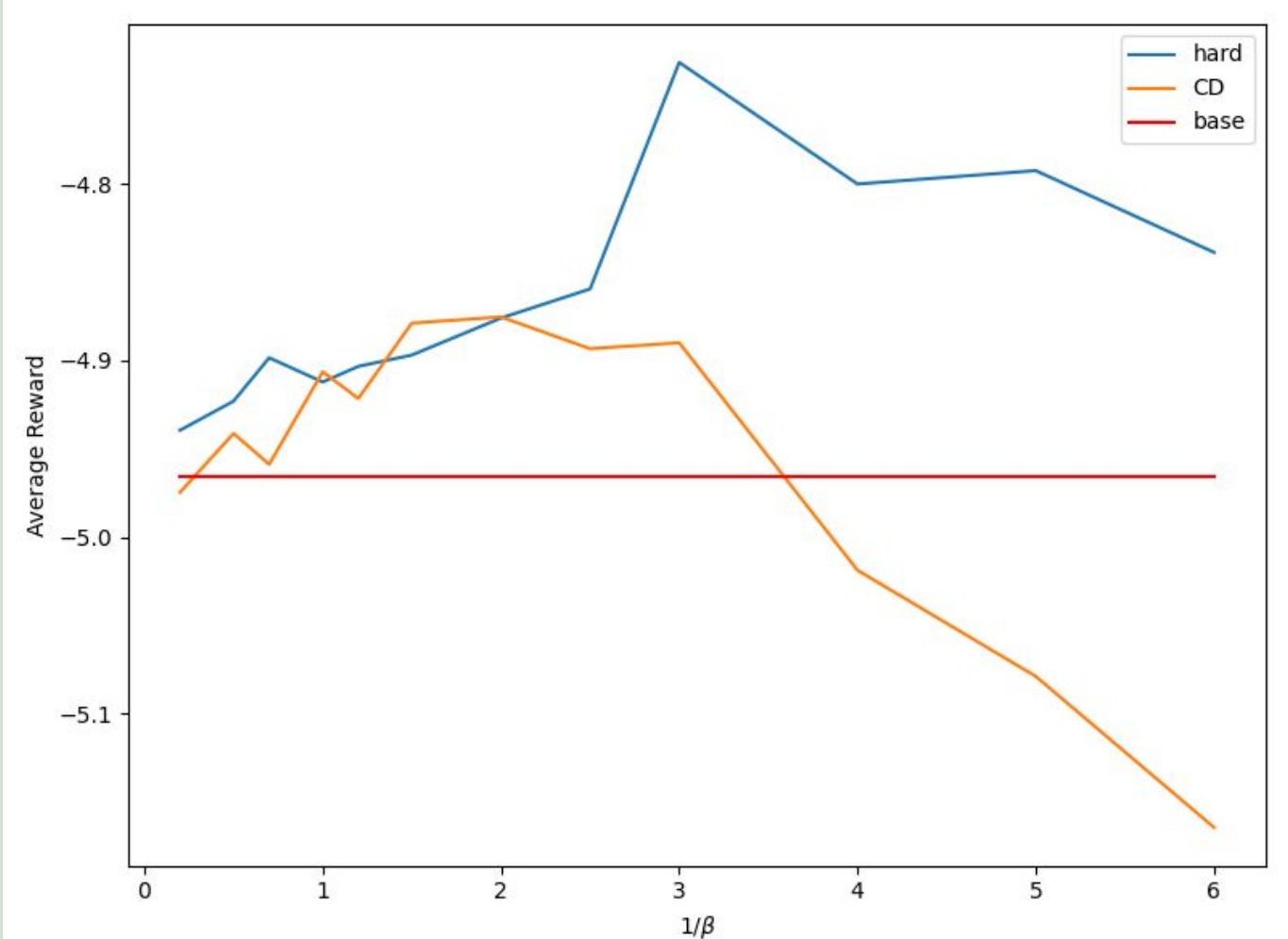
- We formulate Principal – Agent Decoding (PAD): an optimal reward-shaping problem under KL-constrained best response
- We derive closed-form optimal reward structure: prompt-dependent thresholded (0/B) reward
- Provide empirical evidence: higher reward, similar coherence, and robust performance across different levels of regularization



“ Reward shaping before inference-time alignment produces outputs with superior average rewards and higher coherence compared to existing controlled decoding methods.

Example:

- A user wants a neutral answer from a left-leaning model
- Truthfully reporting the reward doesn’t shift the model enough (KL keeps it near baseline)
- User may exaggerate the reward for neutrality to correct the bias



Across varying levels of the KL-divergence penalty ($1/\beta$), Principal-Agent Decoding (blue line) consistently achieves the highest average reward compared to Standard Controlled Decoding (orange line), and the Base Model (red line).

Background: Controlled Decoding

Controlled decoding lets us steer the model’s output at test time without retraining the model. It guides the model to generate better responses by adjusting token probabilities using a reward function.

1. A baseline model gives a probability for each next token
2. We also have a reward model that scores complete responses
3. Controlled decoding tries to pick tokens that will eventually lead to high-reward responses

Equation 1: Token-level / trajectory-level controlled decoding objective

$$\rho_r(y | x) = \rho_{BL}(y | x) \frac{\exp\left(\frac{1}{\beta} r(x, y)\right)}{Z_r(x)}$$

Optimal Policy (Best Response) $\rightarrow \rho_r(y | x)$ $\leftarrow \rho_{BL}(y | x)$ Baseline Policy (Prior Distribution) $\leftarrow Z_r(x)$ Exponential Reward Weighting (Likelihood) $\leftarrow \exp\left(\frac{1}{\beta} r(x, y)\right)$ Normalization Constant (Partition Function) $\leftarrow Z_r(x)$

Equation 2: Closed-form best response

$$\rho_r(\cdot | x) = \arg \max_{\rho} \mathbb{E}_{y \sim \rho(\cdot | x)} [Q(x, y)] - \beta \cdot D_{KL}(\rho(\cdot | x) \| \rho_{BL}(\cdot | x))$$

Optimal Policy Distribution $\rightarrow \rho_r(\cdot | x)$ Optimization over all policies $\rightarrow \arg \max_{\rho}$ Expected Reward (Utility) $\rightarrow \mathbb{E}_{y \sim \rho(\cdot | x)} [Q(x, y)]$ Regularization Coefficient (Inverse Temperature) $\rightarrow \beta$ KL Divergence (Distance from Baseline Policy) $\rightarrow D_{KL}(\rho(\cdot | x) \| \rho_{BL}(\cdot | x))$ Baseline Policy Distribution $\rightarrow \rho_{BL}(\cdot | x)$

Controlled Decoding Ignores Incentives

Controlled decoding assumes the user (or platform) will honestly report their reward function. But this isn’t true in general:

- Users can benefit from exaggerating the reward of answers they prefer
- Because the model is kept close to the baseline, truthful rewards might not move the model enough so users have incentives to distort the reward scale

Our Approach: Principal-Agent Decoding

We view decoding from the lens of principal-agent model.

- **Principal:** The entity who designs the incentive scheme (reward function)
- **Agent:** The LM which best responds in response to the reward model

Principal-Agent Decoding (PAD) answers the core question:

“What reward function should the principal submit to get the best aligned behavior?”

PAD proposes the following optimization problem motivated by alignment objectives:

$$\begin{aligned} \max_{\rho} \quad & \mathbb{E}_{y \sim \rho_r} [r_P(x, y)] \\ \text{s.t.} \quad & \rho_r(\cdot | x) = \arg \max_{\rho} \mathbb{E}_{y \sim \rho(\cdot | x)} [r(x, y)] - \beta \cdot D_{KL}(\rho(\cdot | x) \| \rho_{BL}(\cdot | x)) \\ & 0 \leq r(x, y) \leq B, \forall y \end{aligned}$$

We derive the optimal reward shaping:

$$r^*(x, y) = \begin{cases} B, & \text{if } r_P(x, y) \geq m^*(x), \\ 0, & \text{otherwise.} \end{cases}$$

where $m^*(x)$ is a prompt-dependent cutoff, for which we show can be approximated by a Monte Carlo estimator.

Main Experiment

We applied 3 decoding methods: baseline, standard controlled decoding, and PAD and evaluated their responses on the HH-RLHF alignment dataset using the specified metrics.

Algorithm

Step 1: Perform reward shaping

Step 2: Use the shaped reward in standard controlled decoding procedure

Setup

- **Dataset:** HH-RLHF
- **Base model:** Llama-3 8B Instruct.
- **Reward model:** UltraRM-13B (public RM).
- **Metrics:** Diversity, Coherence (SimCSE cosine), Avg reward, GPT-4 win–tie vs dataset-preferred responses

Results

Dataset	Backbone	Model	Diversity $\uparrow$	Coherence $\uparrow$	Avg Reward $\uparrow$	Win-Tie Rate (%) $\uparrow$
HH-RLHF	Llama3-8B	Base	0.808	0.589	-4.97	78
		CD (prefix)	0.796	0.607	-4.89	83
		PAD	0.787	0.593	-4.73	84

- Our method achieves roughly twice the reward gain obtained by controlled decoding while preserving output coherence
- Our method augments existing controlled decoding techniques, incurring no computation overhead.
- Diversity drops slightly (expected for inference-time alignment)
- Win–tie rate improves only marginally because the base model is already highly aligned